Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 1 of 16 · Question 7 part a.i · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

7	(a) (i)	• when both present; • both (equally) expressed;		2
	(ii)	• I^BI^O; • I^BI^B;		2
	(b)	any four from • carried on X chromosome; • not expressed in heterozygotes/recessive; • can't have male carriers; • can only have female heterozygotes/carriers; • because only females can have 2X chromosomes • male XY/only one X chromosome;		4

Total 8 marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 1 of 16 · Question 7 part a.i · 2 marks

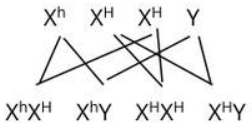
2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 2 of 16 · Question 9 part a.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

9	(a)	(i)	A (the inheritance of two different alleles, both of which are expressed) B is incorrect as one allele is not expressed in codominant inheritance C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles		1
		(ii)	I^AI^A and I^AI^o		1
	(b)	(i)	• blood group O contains no antigens; • antigens removed/digested (by enzyme); • carbohydrase digest carbohydrates; • protease digest proteins;		Max 3
		(ii)	reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection;		1
	(c)	(i)	parent genotypes X^hX^H and X^HY; gametes and linkage X^h X^H X^H Y  ;	One mark for parent genotypes and one mark for gametes **and** linkage	1 1
		(ii)	1 in 4/25%/quarter/1/4 /1:3/0.25;		1

Total question 9 = 9 mark

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 2 of 16 · Question 9 part a.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 3 of 16 · Question 8 part a.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
8 (a)	(i)	B; (an alternative form of gene) A it is not an allele C it is not an allele D it is not an allele		1
	(ii)	only expressed in the homozygous state/not expressed in the heterozygous state;	Allow valid descriptions/examples of homozygous/heterozygous states	1
(b)	(i)	grandfather (condition) × grandmother; (no condition) father (cond) × mother (no cond) ; brother (condition) sister (no condition); student (condition)		3
	(ii)	• occurs only in males/not females/more common in males/females are carriers; • recessive; • probably sex-linked/carried on X chromosome;		3 Total 8

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 3 of 16 · Question 8 part a.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 4 of 16 · Question 8 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
8				
(a)	(i)	B; (an alternative form of gene) A it is not an allele C it is not an allele D it is not an allele		1
	(ii)	only expressed in the homozygous state/not expressed in the heterozygous state;	Allow valid descriptions/examples of homozygous/heterozygous states	1
(b)	(i)	grandfather (condition) × grandmother; (no condition) father (cond) × mother (no cond) ; brother (condition) sister (no condition); student (condition)		3
	(ii)	• occurs only in males/not females/more common in males/females are carriers; • recessive; • probably sex-linked/carried on X chromosome;		3 Total 8

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 4 of 16 · Question 8 part a.ii · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 5 of 16 · Question 5 part a · 2 marks

2023 NOV · Paper 02 · find part a in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 6 of 16 · Question 3 part a.ii · 1 mark

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

3 (a) (i)	nucleus;			(1)
(ii)	allele;			(1)
(iii)	too many fingers/toes;			(1)
(b) (i)	heterozygous/heterozygote/carrier;			(1)
(ii)	Y;			(1)
(c)	Any four from: • double helix/two strands; • consists of nucleotides; • four bases; • each strand held together by hydrogen bonds/bases linked by hydrogen bonds; • complementary base pairing/A-T, G-C;			(4)

Total Question 3 = 9 marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 7 of 16 · Question 3 part b.i · 1 mark

2024 JUN · Paper 01 · find part b.i in this mark scheme extract

3 (a)	(i)	nucleus;		(1)
	(ii)	allele;		(1)
	(iii)	too many fingers/toes;		(1)
(b)	(i)	heterozygous/heterozygote/carrier;		(1)
	(ii)	Y;		(1)
(c)		Any four from: • double helix/two strands; • consists of nucleotides; • four bases; • each strand held together by hydrogen bonds/bases linked by hydrogen bonds; • complementary base pairing/A-T, G-C;		(4)

Total Question 3 = 9 marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 8 of 16 · Question 4 part b.i · 1 mark

2024 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

4 (a)	(i)	11, 31, 30		(1)
	(ii)	31 (× 100); 100 = 31%;	send to review if other than 31	(2)
(b)	(i)	**D (alternative forms of the same gene in a genotype)** The following are incorrect because: A two dominant genes are the same type of allele B two recessive genes are the same type of allele C alleles are not different genes		(1)
	(ii)	• sperm and egg/gametes each contain one allele/haploid; • sperm and egg/gametes fuse; • during fertilisation; • (fertilisation) brings alleles together (in diploid cell)/chromosomes (from each gamete) are paired/one allele comes from each parent;		(4)
(c)	(i)	Hh/heterozygous; hh/homozygous recessive;	either order any suitable letters to be used for alleles	(2)
	(ii)	wheezing/coughing/shortness of breath/low weight / fatigue/over production of mucus/diarrhoea/male infertility;	allow other valid symptoms	(1)

Total for Question 4 = 11 marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 9 of 16 · Question 6 part a · 1 mark

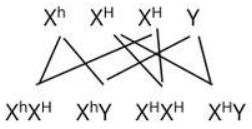
2025 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

6	(a)		An observable characteristic/trait of an individual		1
	(b)	(i)	any two from • caused by a mutation; • (giving a) recessive allele; • both/two alleles need to be present/homozygous recessive genotype;		2
		(ii)	any four from • produces/build up of (thick sticky) mucus (on membranes); • in lungs/small intestine/pancreas/sperm ducts; • coughing/difficulty breathing/less gas exchange/oxygen absorbed; • reduced (aerobic) respiration/less energy; • reduced digestion/ absorption of nutrients; • infertility in males;		4
		(iii)	• some people are carriers/have a heterozygous genotype; • one allele passed on from each parent;		2
				Total for Question 6 = 9 marks	

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 10 of 16 · Question 9 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

9	(a)	(i)	A (the inheritance of two different alleles, both of which are expressed) B is incorrect as one allele is not expressed in codominant inheritance C is incorrect as the ABO inheritance does not involve the inheritance of multiple alleles D is incorrect as ABO inheritances does not involve the inheritance of multiple alleles		1
		(ii)	I^AI^A and I^AI^o		1
	(b)	(i)	• blood group O contains no antigens; • antigens removed/digested (by enzyme); • carbohydrase digest carbohydrates; • protease digest proteins;		Max 3
		(ii)	reduces need to find donor/cheaper/blood readily available/less chance of transfusing incorrect blood group/less risk of disease/infection;		1
	(c)	(i)	parent genotypes X^hX^H and X^HY; gametes and linkage X^h X^H X^H Y  	One mark for parent genotypes and one mark for gametes **and** linkage	1 1
		(ii)	1 in 4/25%/quarter/1/4 /1:3/0.25;		1

Total question 9 = 9 mark

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 10 of 16 · Question 9 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

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Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 11 of 16 · Question 7 part a.ii · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
7	(a)	(i)	• when both present; • both (equally) expressed;		2
		(ii)	• I^BI^O; • I^BI^B;		2
	(b)		any four from • carried on X chromosome; • not expressed in heterozygotes/recessive; • can't have male carriers; • can only have female heterozygotes/carriers; • because only females can have 2X chromosomes • male XY/only one X chromosome;		4

Total 8 marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 11 of 16 · Question 7 part a.ii · 2 marks

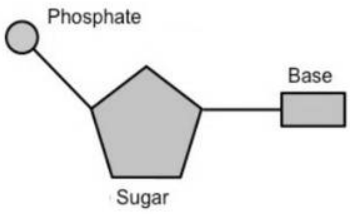
2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 12 of 16 · Question 8 part a.ii · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

8 (a) (i)	Any two from: • (movement of substances/named solute or gas) against a concentration gradient(1) • using energy/ATP(1)		2
(ii)	Any two from: • parent homozygous recessive/heterozygous / carriers (1) • one (recessive) allele inherited from each parent (1) • reference to homozygous recessive disorder/caused by recessive alleles (1)		2
(iii)	• protein/CFTR is not modified/folded/protein has an incorrect shape(1) • protein/CFTR not transported (to its destination)(1)		2
(iv)	1408 x 3 (1) 4224	Full marks for correct final answer. Max 2 marks ecf for final answer if it is correct from calculation	2
(v)	• (deoxyribose) sugar(1) • joined to a (nitrogenous) base/named base(1) • phosphate group attached to sugar (1)  	Allow one mark only if all 3 components just listed Fully labelled diagram for full marks	3
(b)	any three from • deletion(1)	allow description of deletion	3

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 12 of 16 · Question 8 part a.ii · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

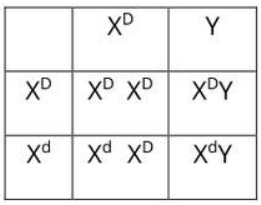
	• of one codon/TTT (1) • phenylalanine missing from protein (1) • shape of final protein changed/different protein formed (1)		

Total for Question 8 = 14 Marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 13 of 16 · Question 8 part b.i · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

8 (a)	any 4 from • change in base sequence produces different mRNA; • altered codon (or codons); • so different amino acids coded; • incorrect protein made; • could produce a stop codon so no protein made/faulty protein made;		4
(b) (i)	• faulty allele present on one chromosome only; • effects overridden by a dominant allele;		2
(ii)	• one male in generation 1 has DMD; • (faulty) allele must be on X chromosome; • male is XY/does not carry dominant allele on Y chromosome; • carriers present;		3
(iii)	• correct parental genotypes; • correct genotypes for generation 1; 		2
			Total 11

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 13 of 16 · Question 8 part b.i · 2 marks

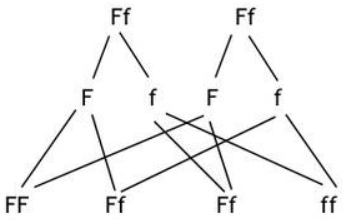
2022 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 14 of 16 · Question 8 part a.iii · 2 marks

2024 JUN · Paper 02 · find part a.iii in this mark scheme extract

Number				
8 (a)	(i)	**B** individual 3 The following are incorrect because: **A** individual 1 is heterozygous **C** Individual 5 is heterozygous **D** individual 3 is homozygous dominant		(1)
	(ii)	**D** 4 The following are incorrect because: **A** there are more than 1 heterozygous genotypes **B** there are more than 2 heterozygous genotypes **C** there are more than 3 heterozygous genotypes		(1)
	(iii)	• they have a dominant allele; • over-rides/ masks the effect of the recessive allele/ must have two recessive alleles for the condition to show/ must have homozygous recessive alleles ;	do not allow dominant allele overpowering the recessive allele	(2)
	(iv)	• heterozygous parent genotypes; • gametes; • genotypes of offspring; • offspring genotypes linked to the phenotype for the disease and showing that showing that 1 of the 4 genotype has the disease;  	mps 2,3 and 4 marks for correct Punnett square showing all details	(4)
	(b)	• reference to use of restriction enzyme • (to obtain) the normal/ non faulty/ dominant allele); • use of virus/ correctly named virus; • inserts allele into person **cells**; • functional/normal protein produced;	Accept liposomes ignore plasmids Penalise one for using genes instead of alleles.	(3)

Total for Question 8 = 11 marks

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 15 of 16 · Question 6 part b.i · 2 marks

2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

6	(a)	An observable characteristic/trait of an individual		1
	(b) (i)	any two from • caused by a mutation; • (giving a) recessive allele; • both/two alleles need to be present/homozygous recessive genotype;		2
	(ii)	any four from • produces/build up of (thick sticky) mucus (on membranes); • in lungs/small intestine/pancreas/sperm ducts; • coughing/difficulty breathing/less gas exchange/oxygen absorbed; • reduced (aerobic) respiration/less energy; • reduced digestion/ absorption of nutrients; • infertility in males;		4
	(iii)	• some people are carriers/have a heterozygous genotype; • one allele passed on from each parent;		2
			Total for Question 6 = 9 marks	

Type MS · Use genetic terms: alleles, genotypes, carriers and blood-group genotypes

Answer 16 of 16 · Question 6 part b.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

6	(a)	An observable characteristic/trait of an individual		1
	(b) (i)	any two from • caused by a mutation; • (giving a) recessive allele; • both/two alleles need to be present/homozygous recessive genotype;		2
	(ii)	any four from • produces/build up of (thick sticky) mucus (on membranes); • in lungs/small intestine/pancreas/sperm ducts; • coughing/difficulty breathing/less gas exchange/oxygen absorbed; • reduced (aerobic) respiration/less energy; • reduced digestion/ absorption of nutrients; • infertility in males;		4
	(iii)	• some people are carriers/have a heterozygous genotype; • one allele passed on from each parent;		2
			Total for Question 6 = 9 marks	